SDS8102: Advanced Mathematical Statistics

Lecture 6: Likelihood Ratio Tests and Confidence Sets

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Lecture 6: Likelihood Ratio Tests and Confidence Sets

Last lecture, we studied optimal tests under special structure:

- Neyman–Pearson for simple versus simple hypotheses;
- monotone likelihood ratio and one-sided UMP tests;
- unbiasedness and UMPU tests for some two-sided problems.

Today we develop tools that apply more broadly:

$$\boxed{\text{GLRT} \to p\text{-values} \to \text{multiple testing} \to \text{confidence sets.}}$$

Learning Objectives

By the end of this lecture, you should be able to

1. Construct a generalized likelihood ratio test.
2. Handle nuisance parameters in Gaussian testing.
3. Define and interpret a p-value.
4. Explain the multiple testing problem.
5. Apply Bonferroni and Benjamini–Hochberg procedures.
6. Construct confidence intervals and regions.
7. Explain test–confidence set duality.

Beyond Simple Hypotheses

Recall the Neyman–Pearson problem:

$$H_0 : \theta = \theta_0 \qquad \text{versus} \qquad H_1 : \theta = \theta_1.$$

We compare two likelihoods:

$$L(\theta_0; x) \qquad \text{and} \qquad L(\theta_1; x).$$

But many problems have the form

$$H_0 : \theta \in \Theta_0 \qquad \text{versus} \qquad H_1 : \theta \notin \Theta_0.$$

Now there are many possible likelihoods under each hypothesis.

$$\boxed{\text{Which likelihoods should we compare?}}$$

Best Constrained versus Best Unrestricted Fit

Under H_0, find the parameter that best explains the data:

$$\widehat{\theta}_0 = \arg\max_{\theta \in \Theta_0} L(\theta; x).$$

Without imposing the null:

$$\widehat{\theta} = \arg\max_{\theta \in \Theta} L(\theta; x).$$

We then compare

$$\boxed{L(\widehat{\theta}_0; x) \qquad \text{versus} \qquad L(\widehat{\theta}; x).}$$

If forcing $\theta \in \Theta_0$ causes a substantial loss in likelihood, the data provide evidence against H_0.

Generalized Likelihood Ratio

Definition: Generalized Likelihood Ratio

The generalized likelihood ratio is

$$\boxed{\Lambda(x) = \frac{\sup\limits_{\theta \in \Theta_0} L(\theta; x)}{\sup\limits_{\theta \in \Theta} L(\theta; x)}.}$$

Since $\Theta_0 \subseteq \Theta$,

$$0 \leq \Lambda(x) \leq 1.$$

Thus

- $\Lambda(x) \approx 1$: the null fits nearly as well as the full model;
- $\Lambda(x) \ll 1$: the null imposes a substantial loss of fit.

We reject H_0 for **small** values of $\Lambda(X)$.

The Likelihood Ratio Test

A generalized likelihood ratio test rejects when

$$\Lambda(X) \leq c.$$

It is often more convenient to work with

$$\boxed{-2 \log \Lambda(X).}$$

Since $-\log(\cdot)$ reverses the ordering,

$$\Lambda(X) \text{ small} \quad \Longleftrightarrow \quad -2 \log \Lambda(X) \text{ large.}$$

Therefore the rejection region can equivalently be written as

$$-2 \log \Lambda(X) \geq c'.$$

The critical value is chosen to control the Type I error.

Example: Gaussian Mean

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad \sigma^2 \text{ known},$$

and consider

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu \neq \mu_0.$$

The unrestricted MLE is

$$\widehat{\mu} = \overline{X},$$

while under H_0,

$$\widehat{\mu}_0 = \mu_0.$$

Therefore

$$\Lambda(X) = \frac{L(\mu_0; X)}{L(\overline{X}; X)}.$$

The GLRT Recovers the z-Test

Using

$$\sum_{i=1}^{n}(X_i-\mu_0)^2=\sum_{i=1}^{n}(X_i-\overline{X})^2+n(\overline{X}-\mu_0)^2,$$

we obtain

$$\boxed{-2\log\Lambda(X)=\frac{n(\overline{X}-\mu_0)^2}{\sigma^2}.}$$

Thus the GLRT rejects for large values of

$$\left|\frac{\sqrt{n}(\overline{X}-\mu_0)}{\sigma}\right|.$$

Under H_0 this statistic is standard Normal, so the level-α test rejects when

$$\boxed{\left|\frac{\sqrt{n}(\overline{X}-\mu_0)}{\sigma}\right|>z_{1-\alpha/2}.}$$

From Decisions to Strength of Evidence

A level-α test produces a binary decision:

$$\text{reject} \qquad \text{or} \qquad \text{do not reject.}$$

Suppose two experiments give

$$|Z| = 2.01 \qquad \text{and} \qquad |Z| = 5.20.$$

At level $\alpha = 0.05$, both lead to rejection.

But the second observation is much more extreme under H_0.

Can we report **how incompatible** the observed statistic is with the null, rather than only a binary decision?

The p-Value

Suppose large values of a test statistic T provide evidence against H_0.

For observed data x_{obs}, define

$$\boxed{p = P_{H_0}\left(T(X) \geq T(x_{\mathrm{obs}})\right).}$$

Thus the p-value measures how extreme the observed test statistic is relative to its null distribution.

A small p-value means that observations at least this extreme would be unusual under H_0.

p-Values and Significance Levels

An equivalent interpretation is

$$\boxed{p(x) = \inf \{\alpha : H_0 \text{ would be rejected at level } \alpha\}.}$$

Therefore,

$$\boxed{p \leq \alpha \quad \iff \quad \text{reject } H_0 \text{ at level } \alpha.}$$

For example, if

$$p = 0.03,$$

we reject at level 0.05, but not at level 0.01.

Thus the p-value summarizes the outcome across different significance levels.

Example: Two-Sided Gaussian Test

For

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu \neq \mu_0,$$

suppose σ is known and we observe

$$z_{\text{obs}} = \frac{\sqrt{n}(\overline{x} - \mu_0)}{\sigma}.$$

Under H_0,

$$Z \sim N(0, 1).$$

Values far from zero in either direction provide evidence against H_0, so

$$\boxed{p = P(|Z| \geq |z_{\text{obs}}|) = 2\left\{1 - \Phi(|z_{\text{obs}}|)\right\}.}$$

For a t-test, the same idea applies using the t_{n-1} distribution.

What a p-Value Is Not

A p-value is **not**

$$P(H_0 \text{ is true} \mid X).$$

It is also not the probability that the observed result occurred "by chance."

Rather, it is calculated **assuming the null is true**:

$$\boxed{P_{H_0}\left(\text{a test statistic at least as extreme as observed}\right).}$$

Thus

$$p = 0.03$$

does **not** mean that there is a 3% probability that H_0 is true.

From One Hypothesis to Many

Modern datasets often involve testing many hypotheses simultaneously.

Examples include:

- Which genes are associated with a disease?
- Which features are associated with an outcome?
- Which regions of an image show a significant effect?
- Which of many experimental interventions have an effect?

Suppose we test

$$H_{01}, H_{02}, \ldots, H_{0m}$$

and perform each test at level $\alpha = 0.05$.

Is the probability of making a false discovery still controlled at 5%?

The Multiple Testing Problem

Suppose all m null hypotheses are true and the tests are independent.

If each test has Type I error probability α, then

$$P(\text{no false rejection}) = (1-\alpha)^m.$$

Therefore,

$$P(\text{at least one false rejection}) = 1-(1-\alpha)^m.$$

For $\alpha = 0.05$,

m	$P(\text{at least one false rejection})$
1	0.05
10	0.40
20	0.64
100	0.994

Testing many hypotheses separately can therefore produce false discoveries even when every null hypothesis is true.

Family-Wise Error Rate

One way to address multiplicity is to control the probability of making **even one** false rejection.

Let V denote the number of false rejections.

The **family-wise error rate** is

$$\boxed{\text{FWER} = P(V \geq 1).}$$

We would like a procedure satisfying

$$\text{FWER} \leq \alpha.$$

This is a stringent criterion:

$$\boxed{\text{Control the probability of making any false discovery.}}$$

Bonferroni Correction

Suppose we test m hypotheses.

The **Bonferroni procedure** rejects H_{0j} when

$$\boxed{p_j \le \frac{\alpha}{m}.}$$

Why does this control the FWER?

Let A_j be the event that a true null H_{0j} is rejected.

By the union bound,

$$P\left(\bigcup_{j\in\mathcal{H}_0} A_j\right) \le \sum_{j\in\mathcal{H}_0} P(A_j) \le m_0\frac{\alpha}{m} \le \alpha \implies \text{FWER} \le \alpha.,$$

where m_0 is the number of true null hypotheses.

No independence assumption is required.

The Price of Bonferroni

Suppose

$$m = 1000, \qquad \alpha = 0.05.$$

Bonferroni requires

$$p_j \leq \frac{0.05}{1000} = 0.00005.$$

This provides strong protection against false positives.

But when m is large, the threshold can become extremely stringent.

As a result, Bonferroni may have low power and fail to detect many genuine effects.

This motivates a less stringent error criterion when our goal is to make **many discoveries**.

False Discovery Rate

Let

- R = total number of rejected hypotheses;
- V = number of rejected hypotheses whose nulls are actually true.

The **false discovery proportion** is

$$\text{FDP} = \frac{V}{R \vee 1}.$$

The **false discovery rate** is

$$\boxed{\text{FDR} = \mathbb{E}\left[\frac{V}{R \vee 1}\right].}$$

Thus FDR controls the expected **proportion** of false discoveries among the rejected hypotheses.

This is fundamentally different from controlling the probability of making even one false discovery.

Benjamini–Hochberg Procedure

Suppose we have m p-values.

First order them:

$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}.$$

For a desired FDR level q, find

$$\boxed{k = \max\left\{j : p_{(j)} \leq \frac{j}{m} q\right\}.}$$

Then reject

$$\boxed{H_{(1)}, \ldots, H_{(k)}.}$$

The threshold becomes less stringent as j increases.

Under independence, and under certain forms of positive dependence, BH controls the FDR at level at most q.

Example: Benjamini–Hochberg

Suppose $m = 8$ and we observe

j	1	2	3	4	5	6	7	8
$p_{(j)}$	.003	.008	.018	.026	.070	.15	.31	.62

Take

$$q = 0.05.$$

The BH thresholds are

j	1	2	3	4	5	6	7	8
jq/m	.0063	.0125	.0188	.025	.0313	.0375	.0438	.05

The largest j satisfying

$$p_{(j)} \leq \frac{j}{m} q \text{ is } j = 3.$$

Therefore BH rejects the first three hypotheses.

Bonferroni versus Benjamini–Hochberg

The two procedures answer different questions.

	Bonferroni	BH
Controls	FWER	FDR
Goal	Avoid any false rejection	Control false discovery proportion
Threshold	α/m	Adaptive: jq/m
Typical behavior	More conservative	More discoveries

Neither criterion is universally "better."

The appropriate choice depends on the scientific goal:

$$\boxed{\text{How costly are false discoveries relative to missed discoveries?}}$$

From Testing to Estimating Uncertainty

A hypothesis test asks whether a particular value θ_0 is compatible with the data:

$$H_0 : \theta = \theta_0.$$

But often we want to know:

$$\boxed{\text{Which values of } \theta \text{ are reasonably compatible with the observed data?}}$$

A point estimator such as

$$\widehat{\theta}$$

gives only one value.

A **confidence set** quantifies the uncertainty surrounding that estimate.

Confidence Sets

Definition: Confidence Set

A random set $C(X)$ is a $(1-\alpha)$ confidence set for θ if

$$\boxed{P_\theta\{\theta \in C(X)\} \geq 1-\alpha \qquad \text{for every } \theta \in \Theta.}$$

The probability is computed over repeated samples

$$X \sim P_\theta.$$

When θ is one-dimensional, $C(X)$ is often an interval:

$$C(X) = [L(X), U(X)].$$

We then call it a **confidence interval**.

What Is Random?

In frequentist inference,

$$\boxed{\theta \text{ is fixed,} \qquad C(X) \text{ is random.}}$$

Before observing the data,

$$P_\theta\{\theta \in C(X)\} = 0.95$$

means that the procedure covers the true parameter in 95% of repeated experiments.

After observing $X = x$, the interval

$$C(x)$$

is fixed.

Therefore, strictly speaking, a 95% confidence interval does **not** mean

$$P\{\theta \in C(x) \mid X = x\} = 0.95.$$

That would require a probability distribution on θ.

Constructing Confidence Intervals

A common strategy is to find a quantity

$$Q(X, \theta)$$

whose distribution does not depend on the unknown parameter.

Such a quantity is called a **pivot**.

If

$$P_\theta \{a \leq Q(X, \theta) \leq b\} = 1 - \alpha,$$

then we can solve the inequalities for θ.

This converts a probability statement about $Q(X, \theta)$ into a confidence set for θ.

Example: Gaussian Mean with Known Variance

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad \sigma^2 \text{ known.}$$

We know

$$Z = \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \sim N(0, 1).$$

Hence

$$P_\mu \left\{ -z_{1-\alpha/2} \leq \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \leq z_{1-\alpha/2} \right\} = 1 - \alpha.$$

Now solve this inequality for μ.

The Gaussian Confidence Interval

Rearranging,

$$P_\mu \left\{ \overline{X} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right\} = 1 - \alpha.$$

Therefore,

$$\boxed{C(X) = \left[\overline{X} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \ \overline{X} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right].}$$

The interval has the familiar form

$$\boxed{\text{estimate } \pm \text{ critical value} \times \text{standard error.}}$$

What Determines the Width?

The Gaussian confidence interval has width

$$\boxed{2z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.}$$

Therefore:

- larger $n \Rightarrow$ narrower interval;
- larger $\sigma \Rightarrow$ wider interval;
- larger confidence level $\Rightarrow$ wider interval.

For example,

$$z_{0.95} \approx 1.645, \qquad z_{0.975} \approx 1.96, \qquad z_{0.995} \approx 2.576.$$

Higher coverage requires paying the price of greater uncertainty in the reported interval.

Unknown Variance

If σ^2 is unknown, we use

$$T = \frac{\sqrt{n}(\overline{X} - \mu)}{S} \sim t_{n-1}.$$

Therefore,

$$P_\mu \left\{ -t_{n-1,1-\alpha/2} \leq \frac{\sqrt{n}(\overline{X} - \mu)}{S} \leq t_{n-1,1-\alpha/2} \right\} = 1 - \alpha.$$

Solving for μ gives

$$\boxed{\overline{X} \pm t_{n-1,1-\alpha/2} \frac{S}{\sqrt{n}}.}$$

The same t distribution that appeared in hypothesis testing now provides the confidence interval.

Tests and Confidence Intervals Are Connected

Consider testing, for every possible value μ_0,

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu \neq \mu_0.$$

For each μ_0, perform a level-α test.

Now collect all parameter values that are **not rejected**:

$$\boxed{C(X) = \{\mu_0 : H_0 : \mu = \mu_0 \text{ is not rejected}\}.}$$

This set is a $(1 - \alpha)$ confidence set.

Inverting the z-Test

For known σ, the level-α two-sided test rejects $H_0 : \mu = \mu_0$ when

$$\left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{\sigma} \right| > z_{1-\alpha/2}.$$

Therefore μ_0 is **not rejected** when

$$\left| \frac{\sqrt{n}(\overline{X} - \mu_0)}{\sigma} \right| \leq z_{1-\alpha/2}.$$

Solving for μ_0 gives

$$\boxed{\overline{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu_0 \leq \overline{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.}$$

This is exactly the usual $(1-\alpha)$ confidence interval.

The Test–Confidence Set Duality

The general principle is

$$\boxed{C_{1-\alpha}(X) = \{\theta_0 : H_0 : \theta = \theta_0 \text{ is not rejected at level } \alpha\}.}$$

Conversely, given a $(1-\alpha)$ confidence set $C(X)$, we can test

$$H_0 : \theta = \theta_0$$

by rejecting whenever

$$\boxed{\theta_0 \notin C(X).}$$

Thus confidence sets and hypothesis tests are two views of the same inferential procedure.

Why Does the Duality Give Correct Coverage?

Suppose $\phi_{\theta_0}(X)$ is a level-α test of

$$H_0 : \theta = \theta_0.$$

Define

$$C(X) = \{\theta_0 : \phi_{\theta_0}(X) = 0\}.$$

When the true parameter is θ,

$$\begin{aligned} P_\theta\{\theta \in C(X)\} &= P_\theta\{\phi_\theta(X) = 0\} \\ &= 1 - P_\theta\{\phi_\theta(X) = 1\} \\ &\geq 1 - \alpha. \end{aligned}$$

Hence

$$\boxed{P_\theta\{\theta \in C(X)\} \geq 1 - \alpha.}$$

Coverage follows directly from Type I error control.

p-Values and Confidence Intervals

For a two-sided level-α test,

$$\boxed{H_0 : \theta = \theta_0 \text{ is rejected} \quad \Longleftrightarrow \quad \theta_0 \notin C_{1-\alpha}(X).}$$

Since

$$p \leq \alpha \quad \Longleftrightarrow \quad H_0 \text{ is rejected},$$

we also have

$$\boxed{p(\theta_0) \leq \alpha \quad \Longleftrightarrow \quad \theta_0 \notin C_{1-\alpha}(X).}$$

So the same inferential information can often be reported as a p-value, a hypothesis test, or a confidence interval.

One-Sided Confidence Bounds

Sometimes we only need to bound the parameter in one direction.

For

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad \sigma^2 \text{ known},$$

we have

$$P_\mu\left\{\frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \le z_{1-\alpha}\right\} = 1 - \alpha.$$

Rearranging,

$$P_\mu\left\{\mu \ge \overline{X} - z_{1-\alpha}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha.$$

Thus

$$\boxed{L(X) = \overline{X} - z_{1-\alpha}\frac{\sigma}{\sqrt{n}}}$$

is a $(1 - \alpha)$ lower confidence bound for μ.

Confidence Bounds and One-Sided Tests

Consider the one-sided test

$$H_0 : \mu \le \mu_0 \qquad \text{versus} \qquad H_1 : \mu > \mu_0.$$

The level-α z-test rejects when

$$\frac{\sqrt{n}(\overline{X} - \mu_0)}{\sigma} > z_{1-\alpha}.$$

Equivalently,

$$\mu_0 < \overline{X} - z_{1-\alpha}\frac{\sigma}{\sqrt{n}}.$$

Therefore, if

$$L(X) = \overline{X} - z_{1-\alpha}\frac{\sigma}{\sqrt{n}},$$

then

$$\boxed{\text{Reject } H_0 : \mu \le \mu_0 \quad \Longleftrightarrow \quad \mu_0 < L(X).}$$

From Confidence Intervals to Confidence Regions

When the parameter is multidimensional,

$$\theta \in \mathbb{R}^d,$$

an interval is no longer sufficient.

Instead, we construct a random set

$$C(X) \subseteq \mathbb{R}^d$$

satisfying

$$\boxed{P_\theta\{\theta \in C(X)\} \geq 1 - \alpha.}$$

Such a set is called a $(1 - \alpha)$ **confidence region**.

The same principles remain:

- find a statistic with a known distribution;
- construct a probability statement;
- invert it to obtain a region for θ.

Example: Gaussian Confidence Ellipsoid

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N_d(\mu, \Sigma),$$

where Σ is known.

Then

$$\sqrt{n}(\overline{X} - \mu) \sim N_d(0, \Sigma).$$

Therefore,

$$n(\overline{X} - \mu)^\top \Sigma^{-1}(\overline{X} - \mu) \sim \chi^2_d.$$

Hence a $(1 - \alpha)$ confidence region is

$$\boxed{C(X) = \left\{\mu : n(\overline{X} - \mu)^\top \Sigma^{-1}(\overline{X} - \mu) \leq \chi^2_{d,1-\alpha}\right\}.}$$

Geometrically, this is an ellipsoid centered at $\overline{X}$.

The Big Picture

We have seen several ways to describe uncertainty about a parameter.

Hypothesis test	$\longleftrightarrow$	Confidence set
Reject $H_0 : \theta = \theta_0$		$\theta_0 \notin C_{1-\alpha}(X)$
$p(\theta_0) \leq \alpha$		$\theta_0 \notin C_{1-\alpha}(X)$

These are not separate ideas.

They are different ways of expressing how compatible candidate parameter values are with the observed data.

Lecture 6: Takeaways

- The **GLRT** compares the best fit under H_0 with the best unrestricted fit.
- In Gaussian models, GLRT leads naturally to the z- and t-tests.
- A p-**value** measures extremeness relative to the null distribution.
- Testing many hypotheses creates a **multiplicity problem**.
- **Bonferroni** controls FWER, while **Benjamini–Hochberg** controls FDR.
- A $(1-\alpha)$ **confidence set** is a random set with coverage probability at least $1-\alpha$.
- Confidence sets can be obtained by **inverting hypothesis tests**.

$\boxed{\text{Testing and confidence sets are two sides of the same idea.}}$